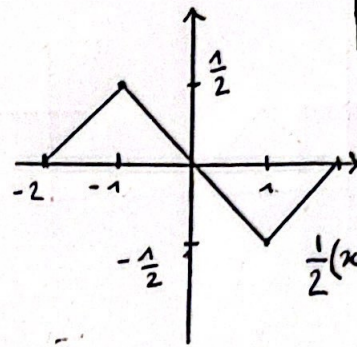
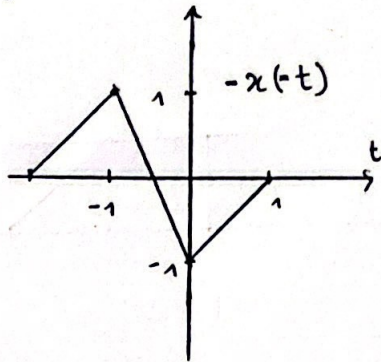
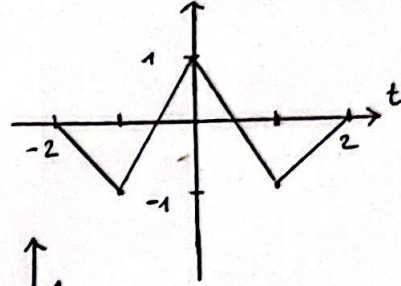
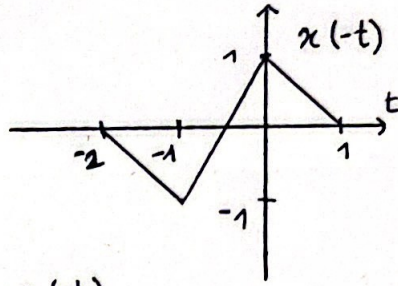
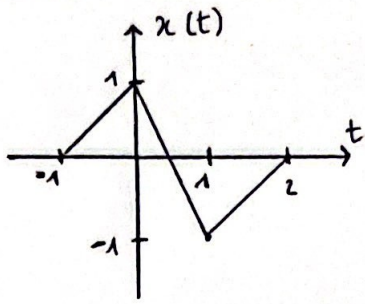


Exercice 1

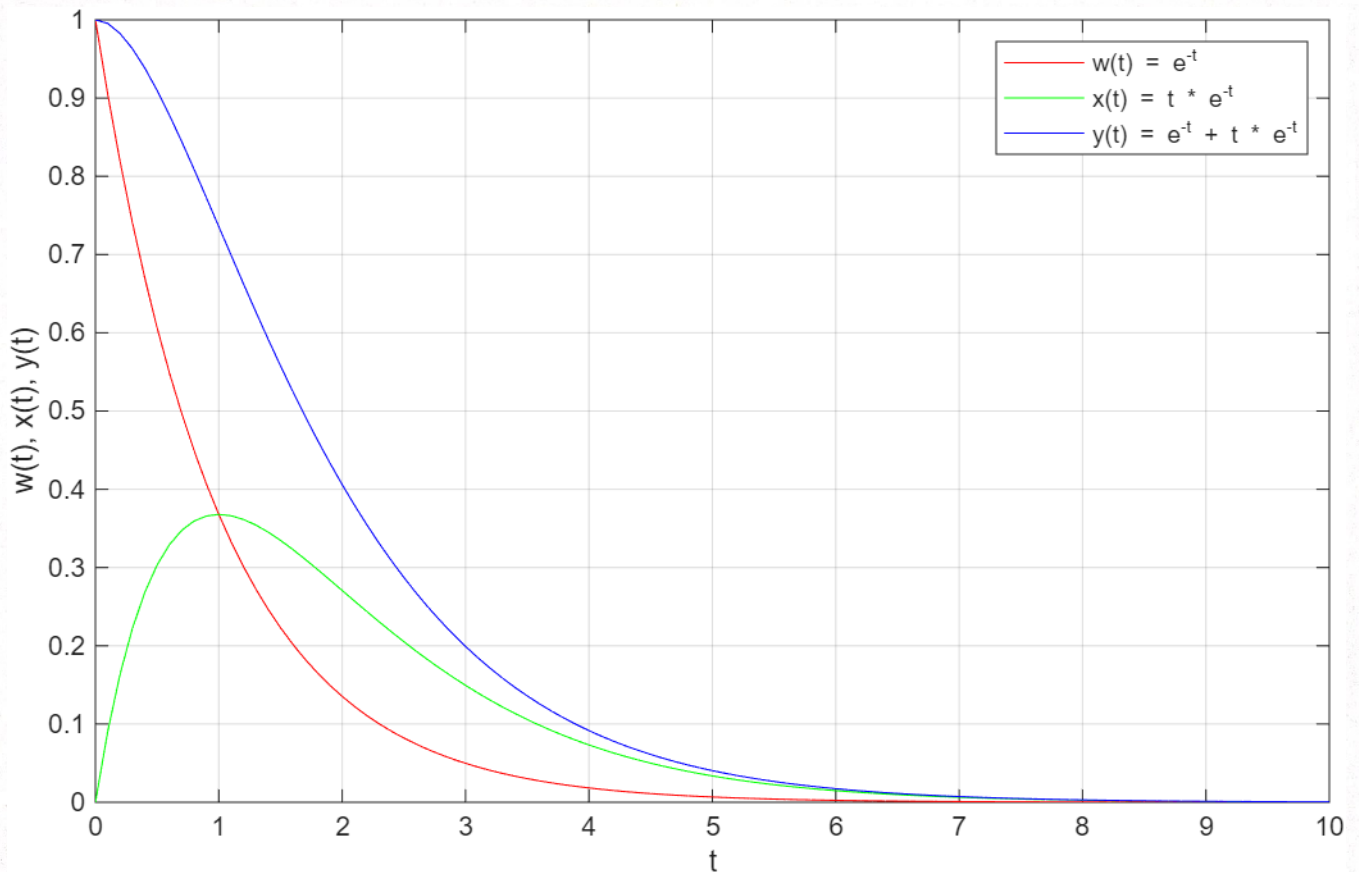
Ines Harraoui

composante paire :
 $\frac{1}{2}(x(t) + x(-t))$



Composante impaire :
 $\frac{1}{2}(x(t) - x(-t))$

Exercice 4



Exercice 2

1) $y(t) = t^2 \frac{dx(t)}{dt}$, $x(t) = x_1(t) + x_2(t)$

- Montrons que $y(t)$ est additif :

D'un côté, on a $y(t) = t^2 \frac{d(x_1(t) + x_2(t))}{dt} = t^2 \left[\frac{dx_1(t)}{dt} + \frac{dx_2(t)}{dt} \right]$

D'un autre côté, on a $y(t) = t^2 \frac{dx_1(t)}{dt} + t^2 \frac{dx_2(t)}{dt} = t^2 \left[\frac{dx_1(t)}{dt} + \frac{dx_2(t)}{dt} \right]$

$t^2 \frac{d(x_1(t) + x_2(t))}{dt} = t^2 \frac{dx_1(t)}{dt} + t^2 \frac{dx_2(t)}{dt}$, donc, $y(t)$ est additif.

- Montrons que $y(t)$ est homogène :

D'un côté, on a $y(t) = t^2 \frac{dx(t)}{dt} \times k$, avec k un gain

D'un autre côté, on a $y(t) = t^2 \frac{d(k \times x(t))}{dt} = t^2 \frac{dx(t)}{dt} \times k$

$t^2 \frac{dx(t)}{dt} \times k = t^2 \frac{d(k \times x(t))}{dt}$, donc, $y(t)$ est homogène et linéaire.

- Montrons que $y(t)$ n'est pas invariant dans le temps :

D'un côté, on a $y(t) = t^2 \frac{dx(t-T)}{dt}$, avec T un retard

D'un autre côté, on a $y(t) = (t-T)^2 \frac{dx(t-T)}{d(t-T)}$

$t^2 \frac{dx(t-T)}{dt} \neq (t-T)^2 \frac{dx(t-T)}{d(t-T)}$, donc, $y(t)$ n'est pas invariant dans le temps.

2) $y(t) = \cos(2\pi f t + x(t))$

- Montrons que $y(t)$ n'est pas additif :

D'un côté, $y(t) = \cos(2\pi f t + x_1(t)) + \cos(2\pi f t + x_2(t))$

D'un autre côté, on a $y(t) = \cos(2\pi f t + (x_1(t) + x_2(t)))$

$\cos(2\pi f t + x_1(t)) + \cos(2\pi f t + x_2(t)) \neq \cos(2\pi f t + (x_1(t) + x_2(t)))$, donc, $y(t)$ n'est pas additif, ni linéaire.

- Montrons que $y(t)$ n'est pas homogène :

D'un côté, $y(t) = \cos(2\pi f t + x(t)) \times k$

D'un autre côté, $y(t) = \cos(2\pi f t + k \times x(t))$

$\cos(2\pi f t + x(t)) \times k \neq \cos(2\pi f t + k \times x(t))$, donc, $y(t)$ n'est pas homogène.

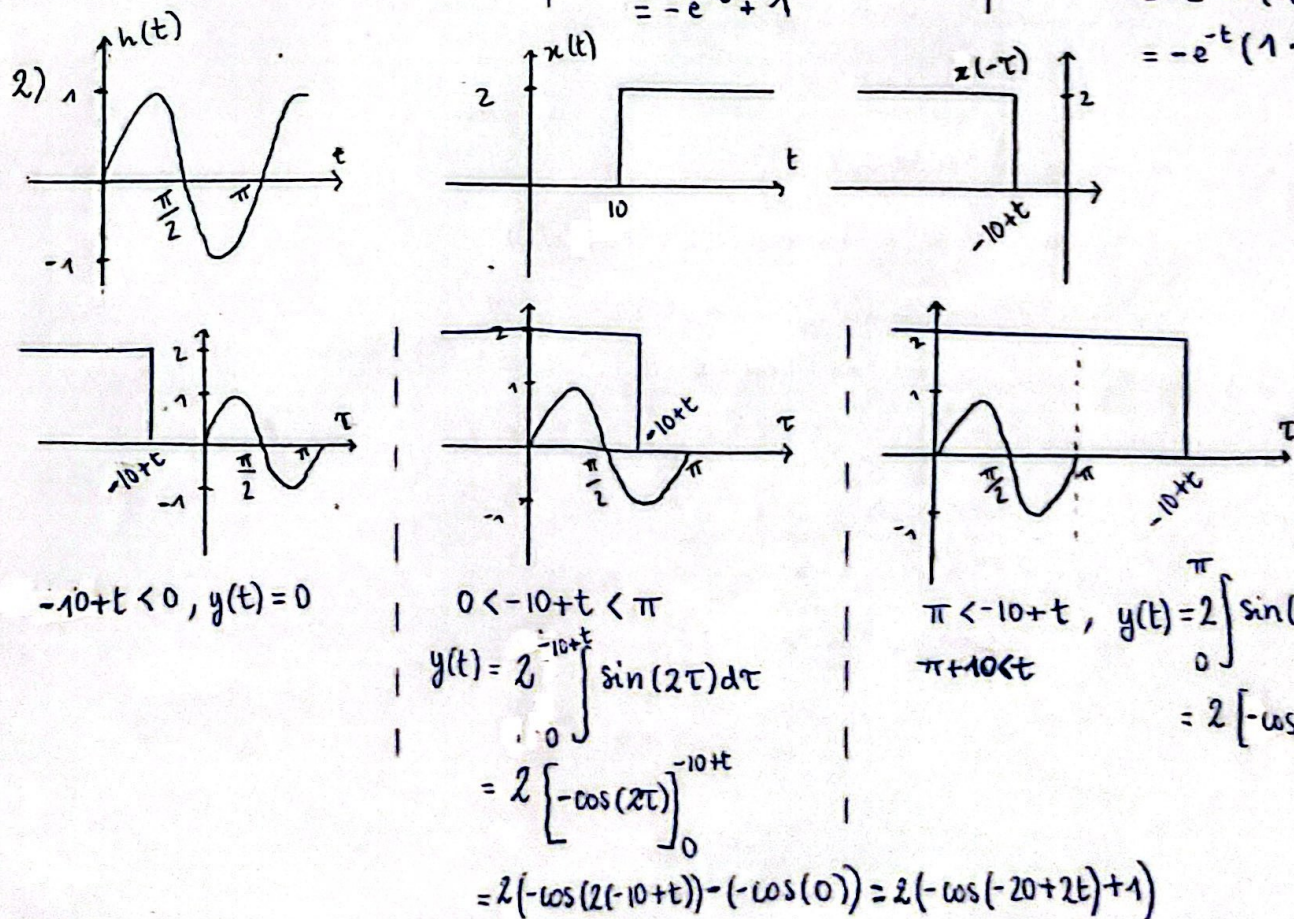
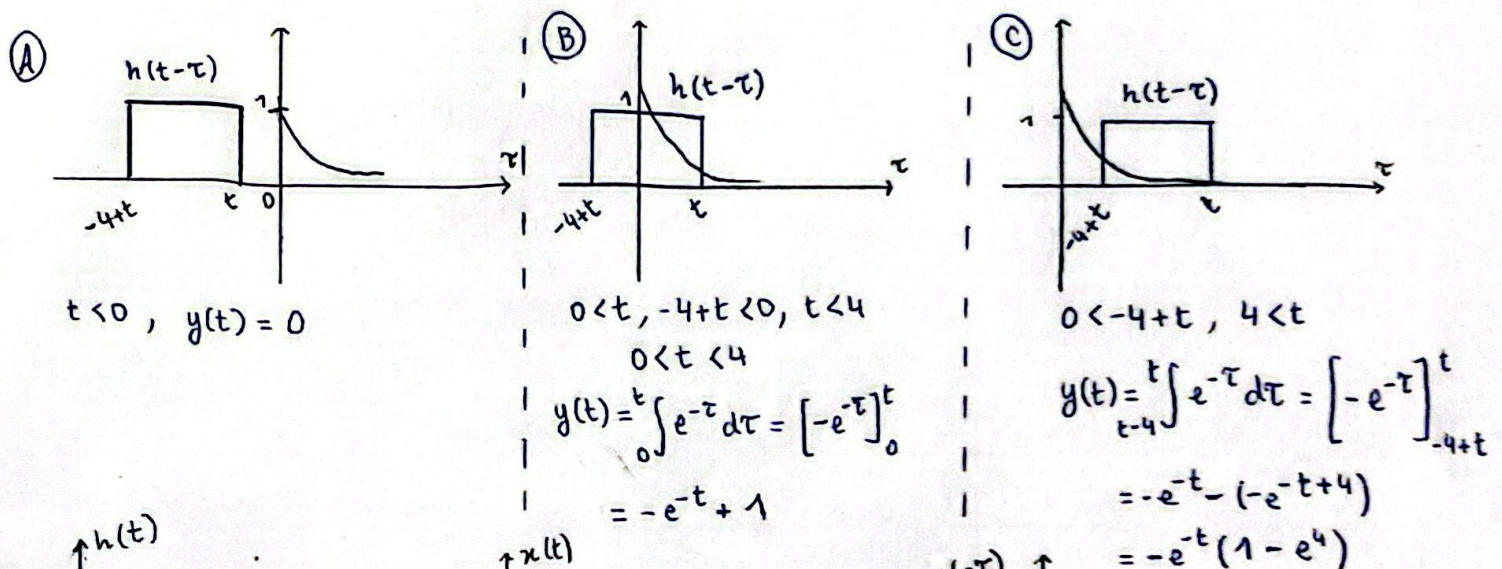
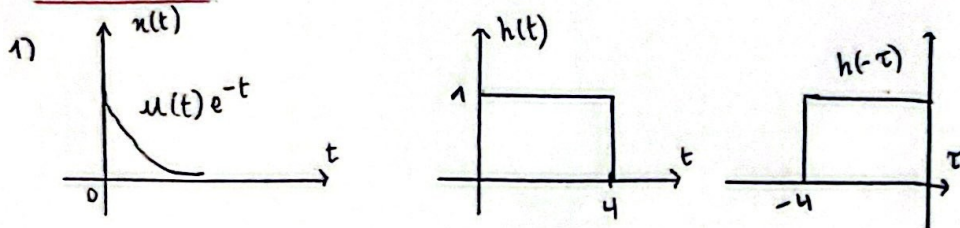
- Montrons que $y(t)$ n'est pas invariant dans le temps

D'un côté, $y(t) = \cos(2\pi f t + x(t-T))$

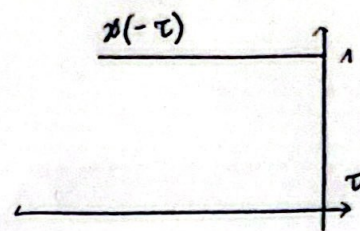
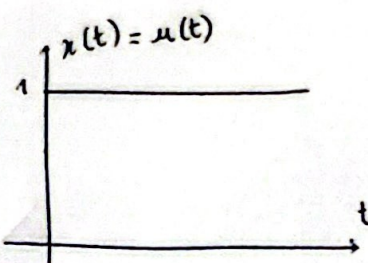
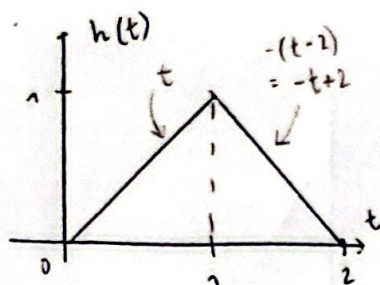
D'un autre côté, $y(t) = \cos(2\pi f(t-T) + x(t-T))$

$\cos(2\pi f t + x(t-T)) \neq \cos(2\pi f(t-T) + x(t-T))$, donc, $y(t)$ est variant dans le temps.

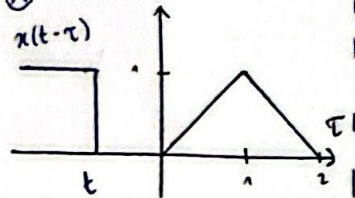
Exercice 3



3)

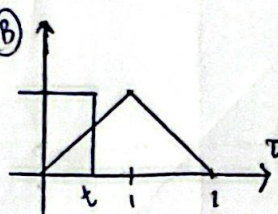


A)



$$t < 0, y(t) = 0$$

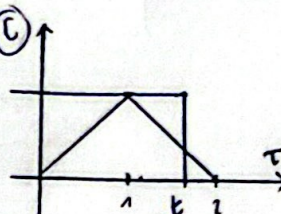
B)



$$0 < t < 1$$

$$y(t) = \int_0^t \tau d\tau = \left[\frac{\tau^2}{2} \right]_0^t = \frac{t^2}{2}$$

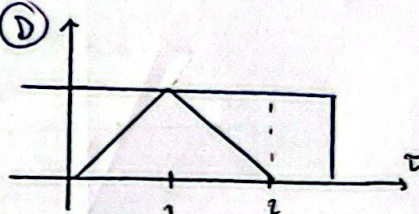
C)



$$1 < t < 2$$

$$y(t) = \int_0^1 \tau d\tau + \int_1^t (2-\tau) d\tau = \frac{1}{2} + \left[2\tau - \frac{\tau^2}{2} \right]_1^t = \frac{1}{2} + 2t - \frac{t^2}{2} - 2 + \frac{1}{2} = 2t - \frac{t^2}{2} - 1$$

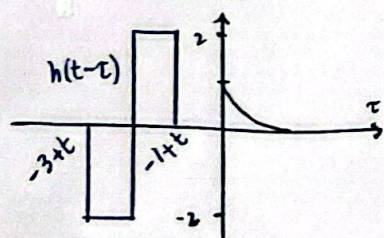
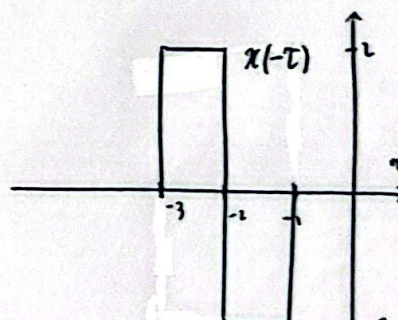
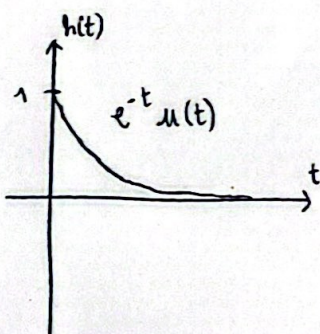
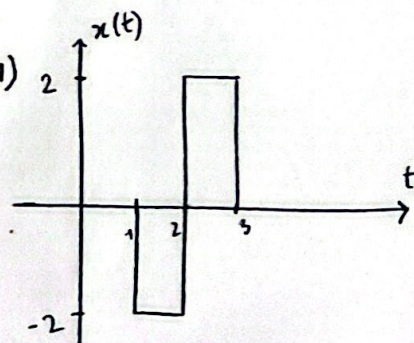
D)



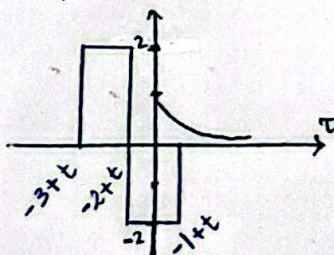
$$t > 2$$

$$y(t) = \int_0^1 \tau d\tau + \int_1^2 (2-\tau) d\tau = \frac{1}{2} + \left[2\tau - \frac{\tau^2}{2} \right]_1^2 = \frac{1}{2} + 4 - 2 - 2 + \frac{1}{2} = 1$$

4)

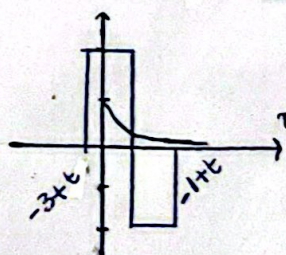


$$-1+t < 0, y(t) = 0$$



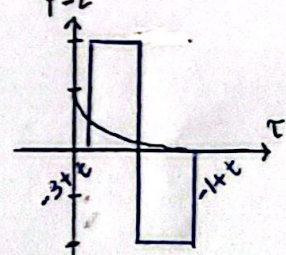
$$-2+t < 0, 0 < -1+t$$

$$y(t) = -2 \int_0^{-1+t} e^{-\tau} d\tau = -2 \left[-e^{-\tau} \right]_0^{-1+t} = -2(-e^{1-t} + 1) = 2e^{1-t} - 2$$



$$0 < -2+t, -3+t < 0$$

$$y(t) = 2 \int_0^{-2+t} e^{-\tau} d\tau + 2 \int_{-2+t}^{-1+t} e^{-\tau} d\tau = 2 \left[-e^{-\tau} \right]_0^{-2+t} - 2 \left[-e^{-\tau} \right]_{-2+t}^{-1+t} = -2(-e^{2-t} + 1) - 2(-e^{1-t} + e^{2-t}) = -2e^{2-t} + 2 + 2e^{1-t} - 2e^{2-t} = -4e^{2-t} + 2 + 2e^{1-t}$$



$$y(t) = 2 \int_0^{-3+t} e^{-\tau} d\tau - 2 \int_{-3+t}^{-1+t} e^{-\tau} d\tau = 2 \left[-e^{-\tau} \right]_0^{-3+t} - 2 \left[-e^{-\tau} \right]_{-3+t}^{-1+t} = 2(-e^{3-t} + 1) - 2(-e^{1-t} + e^{3-t}) = -2e^{3-t} + 2 + 2e^{1-t} - 2e^{3-t} = -4e^{3-t} + 2 + 2e^{1-t}$$